

08/07/26

HW 1) Define a structure in "C" as well as "C++".
This structure should hold the variables for student type. Compute the size manually for this structure and also use `sizeof()` function to display the size. Cross check if both are same or not. Justify your answer.

Self referential structures:

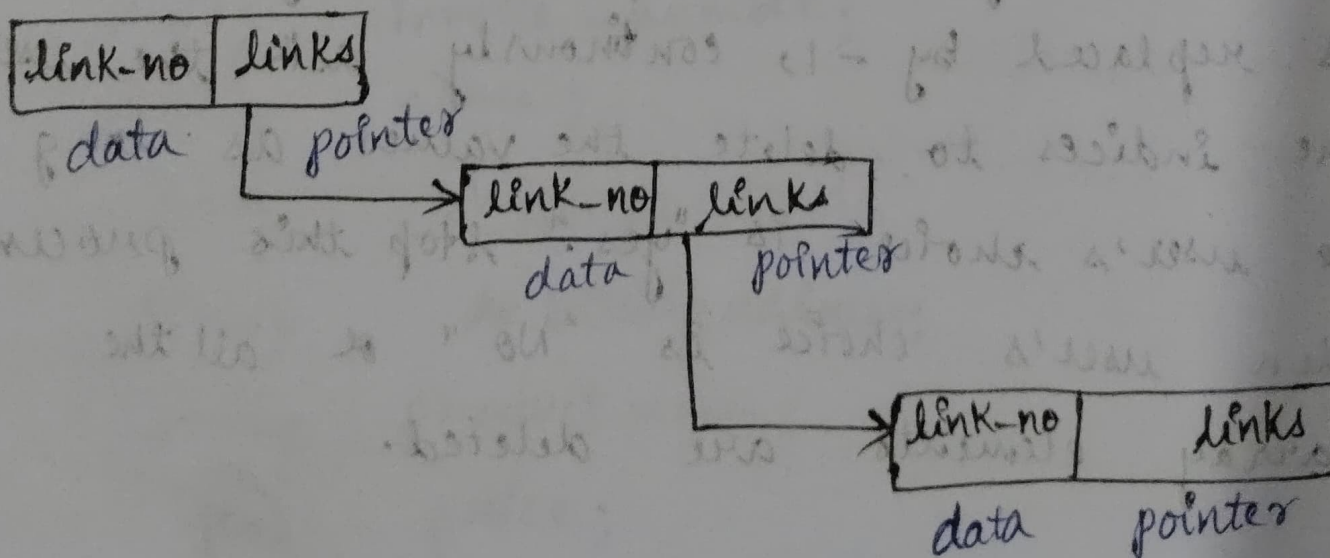
Self referential structures are those structures in which one or more pointers points to the structure of the same type.

```
struct self {
```

```
    int p;
```

```
    struct self *ptr;
```

```
};
```



HW :-

using C

1) #include <stdio.h>

struct student {

int roll;

char grade;

float marks;

};

int main() {

int manual = 4 + 1 + 4;

printf("Manual size = %d bytes \n", manual);

printf("size using sizeof() = %d bytes \n",

sizeof(struct student));

if (manual == sizeof(struct student))

printf("Both are same");

else

printf("Both are not same");

}

output:

Manual size = 9 bytes

Size using sizeof() = 12 bytes

Both are not same.

```

using C++
#include <iostream>
using namespace std;
struct Student {
    int roll;
    char grade;
    float marks;
};

```

```

int main() {
    int manual = 4+1+4;
    cout << "Manual size = " << manual << "bytes" << endl;
    cout << "Size using sizeof() = " << sizeof(struct Student)
    << "bytes" << endl;
}

```

```

if (manual == sizeof(struct Student))

```

```

    cout << "Both are same";

```

```

else

```

```

    cout << "They are not same";
}

```

return 0;

```

}

```

The manual size and the size returned by `sizeof()` is not the same usually because of "structure padding" which is known as "memory alignment".

generally, $\text{int} = 4 \text{ bytes}$
 $\text{char} = 1 \text{ byte}$
 $\text{float} = 4 \text{ bytes}$.

however, the compiler stores like $\text{int} = 4 \text{ bytes}$,
 $\text{char} = 1 \text{ byte}$, padding = 3 bytes, $\text{float} = 4 \text{ bytes}$.

so They are not same because compiler adds padding bytes to align the ~~memory~~ data members in memory. It ~~is~~ improves the performance of memory access.

output :

Manual size = 9 bytes

Size using `sizeof()` = 12 bytes
 They are not same.